The Reliability Tax: Quantifying the Capability–Reliability Trade-off in Reasoning LLMs

Anonymous ACL submission

Abstract

Reasoning-augmented language models (LRMs) are now deployed by default for high-stakes queries, and test-time compute—how long a model is allowed to "think"—is increasingly a user-facing knob. The prevailing assumption is that more thinking yields more trustworthy answers, but recent work has begun to undermine it: sweeping the reasoning budget impairs confidence calibration, and the marginal utility of extra reasoning turns negative. These findings are single-axis and descriptive: they show that the trade-off *exists*, not whether it is uniform or—the property a deployment rule needs—whether it is *predictable*. We ask whether a black-box user can foresee, before committing to a budget, that more thinking will hurt. Our budget-sweep protocol varies thinking across six models from four families over four task types, tracking accuracy and three reliability axes: calibration, selective prediction, and abstention versus fabrication on false-premise items, where *fabrication* means endorsing a premise that is in fact false. We first measure the trade-off and find it is *task-dependent*: **[the ratio of reliability loss to accuracy gain varies by] [x]**× across tasks. We then show it is *partly predictable*: using only the trace and verbalized confidence observed at a low budget, a lightweight predictor decides, on held-out items, whether continuing will help; the resulting oracle-free stopping rule attains higher *usable accuracy*—accuracy at a fixed reliability floor—than the best fixed budget at matched cost, while a permutation control on the trace features loses the advantage. These results recast "think longer" from a default into a decision.

1 Introduction

Reasoning-augmented language models (LRMs) such as o1-style systems and DeepSeek-R1 have become the default for tasks that were previously out of reach, and their defining property is that they spend a variable amount of compute *thinking before answering* (DeepSeek-AI et al., 2025). This has turned test-time compute from an implementation detail into a first-class, increasingly user-facing control: models expose reasoning budgets, "thinking effort" levels, and explicit toggles (Muennighoff et al., 2025; Snell et al., 2024). The implicit contract behind these knobs is simple and appealing: *more thinking should buy more trustworthy answers*. Practitioners act on it—escalating to higher reasoning budgets, longer chains of thought, or a reasoning model over a chat model whenever an answer matters. A customer-support or clinical-QA pipeline, for instance, may simply raise the reasoning budget to "max" for every incoming query, on the assumption that the most careful answer is the safest. The cost of that assumption is concrete: an over-confident wrong answer is worse than an admitted uncertainty, because nothing downstream flags it, and a model that fabricates support for a false premise is more dangerous than one that declines to answer.

This contract is, however, being contradicted from several directions at once. Training that raises reasoning accuracy with verifiable rewards systematically degrades *calibration*: incorrect answers acquire *lower* perplexity than correct ones, so a model's confidence stops tracking its correctness (Wang et al., 2026; Jin et al., 2025). Enhancing reasoning amplifies tool hallucination—models fabricate non-existent tools—and the effect survives even when the enhancement is purely inference-time (Yin et al., 2025). In multimodal settings, "slower" models are measurably *less* truthful under misleading inputs (Fang et al., 2025). Reasoning models commit to false premises more decisively than their non-reasoning counterparts, becoming *confidently wrong* on 26–42% of false-premise queries (Sathyanathan et al., 2026), and even answers that are perfectly self-consistent can

collapse under mild contextual interference (Xu et al., 2026). Most directly, sweeping the reasoning budget itself—at inference time, with no training involved—has been observed, in several settings, to impair confidence calibration (Lacombe et al., 2025) and to yield non-monotonic marginal returns (Zhou et al., 2026).

Each of these is a warning; none is a law. The existing evidence is fragmented along three axes. First, it is *task- and family-local*: a single benchmark, often a single model or training recipe, is used to establish the effect. Second, it is overwhelmingly *training-side*: the sharpest results concern what RLVR does to weights (Wang et al., 2026; Jin et al., 2025), which is unavailable to the many practitioners who only have API access. Third, reliability is measured *one axis at a time*—calibration *or* selective prediction *or* abstention—so it is impossible to tell whether these failures share a cause or merely co-occur. More fundamentally, even the closest work is *descriptive*: none reports whether the three axes degrade *together*, as one underlying cost, or whether the degradation is *predictable* from the reasoning trace before a budget is committed—the property a deployment rule actually needs. Consequently, we cannot currently answer the question that actually governs deployment: *does increasing a model's thinking budget systematically trade reliability for accuracy, and is that trade-off universal or task-specific?*

We address this gap with a controlled, black-box study. We formalize the trade-off as a *reliability tax*: with the thinking budget b as the controlled variable and every reliability score ρ_r oriented so that higher is better, we estimate the ratio of reliability change to accuracy change, $\tau_r = (\partial\rho_r/\partial b)/(\partial \mathrm{acc}/\partial b)$, for each reliability axis r; a large τ_r means accuracy is bought at a steep reliability price—exactly the quantity a deployment should minimize. Because τ_r diverges where accuracy saturates, our primary summaries are bounded instead: the fixed-coverage risk and the area under the risk–coverage curve, with τ_r reported only where $\partial\mathrm{acc}/\partial b > 0$. The three axes correspond to the three decisions a high-stakes deployment must make—*how much to trust an answer* (calibration), *how much coverage to serve* (selective prediction), and *whether to answer at all when the premise is false* (abstention versus fabrication)—and we measure them jointly precisely to test whether they degrade as one or independently. Our *budget-sweep protocol* holds the prompt set fixed and varies a model's thinking budget across its supported range (reasoning-effort levels, reasoning-token caps, and black-box budget forcing (Muennighoff et al., 2025)), tracking, at each budget, accuracy and the three axes: calibration, via ECE and verbalized overconfidence (Kadavath et al., 2022; Tian et al., 2023); selective prediction, via the risk–coverage curve; and the *fabricate-versus-abstain* choice on false-premise items (Sathyanathan et al., 2026; Azin et al., 2026). We sweep six models from four families—open-weight and closed API, all exposing a thinking toggle or a reasoning-effort level—over four task types chosen to span where extra compute should help and where it can hurt: mathematical reasoning (verifiable), factual QA (closed-book knowledge), multi-hop QA (composition), and false-premise items (where the model should refuse). Because budget knobs are not comparable across providers, we report against the *realized* number of reasoning tokens and include an equal-compute control; each item is rendered under several prompt templates so that form-induced variance can be separated from sampling variance (Miller, 2024; Zhang et al., 2025); and all calls are pinned to dated model snapshots, since providers update models silently. Where a provider does not expose raw reasoning traces, our predictor falls back to the answer and verbalized confidence alone, so the protocol never assumes white-box access. We define *usable accuracy* as the accuracy attainable at a fixed reliability floor—the accuracy over the subset served at the coverage where calibration error stays below a target ϵ—so that a budget is judged by what it can be trusted to serve, not by its best-case answer. This design tests one descriptive and two predictive claims: *H1 (task-dependence)* the ratio of reliability loss to accuracy gain differs across task types by at least a factor of two—falsified if the ratios are statistically indistinguishable; *H2 (predictability)* a predictor trained on a held-out half decides, at low budget and without seeing the high-budget answer, whether continuing will help, and the induced policy beats the best fixed budget in usable accuracy at matched cost—falsified otherwise; *H3 (negative control)* shuffling the trace features removes H2's advantage—if the shuffled control matches, the predictor exploits an artifact rather than the trace.

Our contributions are:

- A black-box *budget-sweep protocol* and a multi-axis reliability measurement—calibration, selective prediction, and

abstention-versus-fabrication—that any API user can apply without weights or training.

- To our knowledge, the first cross-task and cross-family test of whether the trade-off is a law or a set of local effects, and of where it is worst.
- A held-out, cost-matched evaluation of an *oracle-free stopping rule* that decides when to stop thinking, together with a permutation control that rules out trace-format artifacts.
- The protocol, prompts, and raw responses, plus a reproducibility recipe that pins dated API snapshots.

2 Related Work

Reasoning models and test-time compute. LRMs are trained to emit long reasoning traces before answering, most prominently through reinforcement learning with verifiable rewards (Shao et al., 2024; DeepSeek-AI et al., 2025). A parallel line studies how to spend *inference-time* compute: scaling test-time search against verifiers can outperform a larger base model (Snell et al., 2024), and simple "budget forcing" can extend or truncate a trace to trade quality for cost (Muennighoff et al., 2025). A complementary strand asks what these traces actually buy. Shojaee et al. (2025) show accuracy collapses past a complexity threshold and that reasoning effort rises then *falls* as problems get harder; Srivastava et al. (2025) find reasoning models generate roughly $18\times$ more tokens while sometimes being *less* accurate, with non-monotonic returns to budget. These works motivate treating the thinking budget as the independent variable, but they evaluate final-answer accuracy rather than reliability, and recent work further questions whether RLVR expands capability at all (Yue et al., 2025; Shao et al., 2025).

Efficient and adaptive reasoning. A large literature instead makes reasoning *cheaper*, by truncating or compressing traces and by early exit (Chen et al., 2024; Sui et al., 2025; Chen et al., 2026; Xiang et al., 2026), while a recent strand characterizes the marginal utility of extra tokens (Zhou et al., 2026). Early-exit methods share our goal of deciding *when to stop*, but differ in both setting and signal: they read the model's *internal* hidden states and are therefore white-box, and they optimize cost subject to accuracy. We stop from the *external* trace and verbalized confidence alone—the only signals an API user has—and we optimize *trust*: the question is not whether the answer stays as accurate, but whether it stays as reliable.

Calibration, selective prediction, and abstention. A well-calibrated model's confidence predicts its correctness, which is what makes deferral and abstention meaningful. Early work showed that conditional probabilities of large pretrained models are reasonably calibrated and that models can verbalize calibrated confidence (Kadavath et al., 2022; Lin et al., 2022), while later work finds that instruction tuning and RLHF harm calibration and that *verbalized* confidence is often the better estimate (Tian et al., 2023; Xiong et al., 2023; Sanz-Guerrero et al., 2026). Most recently, training-side analyses show RLVR makes models overconfident, degrading relative calibration as accuracy rises (Wang et al., 2026; Jin et al., 2025), and that superficially consistent beliefs can be brittle (Xu et al., 2026). The closely related notion of *abstention*—knowing when not to answer—dates back to selective classification (Geifman and El-Yaniv, 2017) and is surveyed for LLMs by Wen et al. (2024); strikingly, reasoning models are *worse* at abstaining on unanswerable questions (Kirichenko et al., 2025). Calibration itself is usually operationalized through ECE, introduced for neural networks by Guo et al. (2017). This literature establishes *that* calibration can degrade with reasoning, but studies one family, one training recipe, or one axis; whether calibration, abstention, and fabrication move together as a function of the budget a black-box user selects is unexamined.

Faithfulness, sycophancy, and false premises. Reasoning traces are not always faithful to the computation that produced the answer (Turpin et al., 2023; Chen et al., 2025), and models are readily steered by user framing or social pressure (Sharma et al., 2023). Hallucination is a growing concern: models fabricate tools as reasoning strengthens (Yin et al., 2025), produce unsupported claims that snowball across steps (Zhang et al., 2023), and cannot reliably self-correct their own reasoning (Huang et al., 2024). On false-premise queries, reasoning models are confidently wrong on a large fraction of items (Sathyanathan et al., 2026), and do not reliably reason about the presuppositions such queries turn on (Azin et al., 2026); multimodal reasoning models become less truthful as they think longer (Fang et al., 2025). Each is typically studied

in isolation; none is probed as a function of the thinking budget.

Reliability pitfalls in evaluation. Measuring reliability is itself error-prone. Benchmark scores carry large sampling variance—on GSM8K a single generation can vary by 18.6 points across runs—so multi-sample, hierarchical estimation is needed (Zhang et al., 2025; Miller, 2024). We therefore adopt multi-sample estimation throughout.

Closest work and our position. The closest studies vary the reasoning budget and measure a single reliability axis: Lacombe et al. (2025) show that over-reasoning impairs confidence calibration, and Zhou et al. (2026) characterize the diminishing, sometimes negative, marginal utility of extra tokens. Both are single-axis, and neither asks whether the losses across axes *co-move* or can be *predicted*. We therefore change the question from *whether* the trade-off exists to *whether it is a law*: we measure calibration, selective prediction, and fabrication jointly across tasks and families, and test whether impending degradation is foreseeable from the trace, turning the trade-off curve into a black-box budget-selection rule.

3 The Reliability Tax

Everything in this paper is read off a single object: the *budget–reliability profile* of a model on a task—how accuracy and each reliability axis change as the model is allowed to think longer. We fix notation and define this object (§3.1), read the tax from it (§3.2), operationalize the three axes (§3.3), and turn the profile into a decision rule (§3.4).

3.1 The budget–reliability profile

Let M be a model and $\mathcal{B} = \{b_0 < \cdots < b_B\}$ its supported thinking budgets—reasoning-effort levels, reasoning-token caps, or black-box budget forcing (Muennighoff et al., 2025). For a prompt x and budget b, the model returns a response $y \sim M(x, b)$ carrying a *realized* reasoning length $\ell(y)$ in tokens. Because a nominal budget means different things to different providers, we plot every quantity against ℓ rather than b. For a fixed prompt set $\mathcal{D}$, the profile is the collection

$$\big\{\mathrm{acc}(b),\ \rho_1(b),\ \rho_2(b),\ \rho_3(b)\big\}_{b\in\mathcal{B}}, \qquad (1)$$

estimated with k samples per (x, b) and reported with confidence intervals clustered by prompt—prompts, not samples, are the unit of variation (Zhang et al., 2025; Miller, 2024). A profile is thus a two-dimensional object, budget against a reliability reading, and the paper is a study of its shape.

3.2 The reliability tax

Orient each reliability score ρ_r so that higher is better, and define the tax along the profile as

$$\tau_r(b) = \frac{\partial \rho_r / \partial b}{\partial \mathrm{acc} / \partial b}, \qquad (2)$$

the reliability given up per unit of accuracy gained. A deployment wants τ_r small; a large τ_r means extra accuracy is bought at a steep reliability price. τ_r is undefined where accuracy saturates—precisely the regime we care about—so our primary summaries are *bounded*: the fixed-coverage risk $R(c)$, the error rate over the top-c fraction of the most confident answers, and the area under the risk–coverage curve (Geifman and El-Yaniv, 2017); we report τ_r only where $\partial \mathrm{acc}/\partial b > 0$.

3.3 Three reliability axes

The axes are the three decisions a high-stakes deployment must make.

Calibration (how much to trust an answer). Expected calibration error (Guo et al., 2017) and *overconfidence*, the mean verbalized confidence on incorrect answers (Kadavath et al., 2022; Tian et al., 2023; Xiong et al., 2023).

Selective prediction (how much coverage to serve). $R(c)$ and AURC, which bound how much of the task a system can answer before its error rate exceeds a target.

Abstention versus fabrication (whether to answer a false premise). On false-premise items we classify each response as *refute*, *hedge*, or *endorse*, and report the fabrication rate—endorsing a premise that is false (Sathyanathan et al., 2026; Kirichenko et al., 2025). The three are measured jointly because the question is not whether any one degrades, but whether they degrade *together*.

3.4 From profile to decision: a stopping rule

The profile also yields an oracle: $b^\star$ maximizes usable accuracy, the accuracy at the coverage where calibration error stays below a target ϵ. The actionable question is whether $b^\star$ can be found *without* running the expensive end of the sweep. We train a predictor on a held-out half of the prompts: at a

low budget b_0 it observes features $f(y)$—realized length, verbalized confidence, self-consistency across k samples, and hedging markers—and decides whether continuing to the top budget will help. The induced policy is evaluated on the other half, compared against every fixed budget under matched cost, and stress-tested by shuffling f across prompts: if the shuffled control matches, the predictor exploits an artifact rather than the trace. Where a provider does not expose raw reasoning traces, f reduces to the answer and its verbalized confidence, so the rule never assumes white-box access.

4 Experimental Setup

Models. Six models from four widely-deployed families, spanning open-weight and closed API, each exposing a thinking toggle or a distinct reasoning variant: DeepSeek (R1 and V3), Qwen3 (235B and 32B, with an `enable_thinking` toggle), GLM-4.6, and Kimi-K2. All are reachable through OpenAI-compatible endpoints, and each provides at least two settings of the budget b, so that b is a real knob rather than a prompt artefact.

Tasks. Four types chosen to span where extra compute should help and where it can hurt: mathematical reasoning (GSM8K, MATH-500; verifiable), factual QA (TriviaQA, NQ; closed-book knowledge), multi-hop QA (HotpotQA; composition), and *false-premise* items, drawn from fact-checked claims known to be false (Wadden et al., 2020) and rewritten as questions that presuppose them (e.g., "Why did Einstein win the 1921 Nobel Prize in Literature?"), where the reliable behaviour is to refute rather than answer. **[200 items per task.]**

Budgets and sampling. **[Four budget levels]** per model, **[** $k = 4$ **]** samples per cell at temperature 0.7, under **[two]** prompt templates so that form-induced variance can be separated from sampling variance (Miller, 2024).

Statistics. Prompt-clustered bootstrap intervals and mixed-effects models with prompt and model as random effects; the negative control is a permutation test over f.

Reproducibility. All calls are pinned to dated model snapshots; prompts, responses, and scripts are released.

4.1 A minimal viable pilot

Before the full sweep we run the smallest end-to-end version that exercises every stage—budget sweep, profile, tax, stopping decision—on a single model and two budgets (**[100 items]**, **[** $k = 4$ **]**). To validate the pipeline itself rather than a model, we first drive it with a synthetic backend that plants a known tax (accuracy rises modestly while calibration and fabrication degrade); the pipeline recovers the planted signs and selects the lower budget as the trustworthy one, and the same script runs unchanged against a real API. The pilot is the unit of validation: if the pipeline cannot recover a known effect here, the larger sweep will not either.

5 Results and Analysis

RQ1: Is the tax real, and is it a law? [Profile shapes per model and task; tax magnitude; and whether the ratio of reliability loss to accuracy gain differs across tasks by the factor of two predicted by H1.]

RQ2: Is it predictable? [Predictor accuracy on held-out prompts; the stopping rule's usable accuracy against the best fixed budget under matched cost (H2); and the shuffled-feature control (H3).]

RQ3: What does it cost in practice? [The share of queries on which the default of "max budget" over-thinks, and the usable accuracy recovered by stopping.]

References

Tara Azin, Yongan Yu, Raj Singh, et al. 2026. Presupposition and reasoning in conditionals: A theory-based study of humans and llms. *Preprint*, arXiv:2605.18352.

Benteng Chen, Weida Wang, Shufei Zhang, et al. 2026. Step-grpo: Internalizing dynamic early exit for efficient reasoning. *Preprint*, arXiv:2604.16890.

Xingyu Chen, Jiahao Xu, Tian Liang, et al. 2024. Do not think that much for 2+3=? on the overthinking of o1-like llms. *Preprint*, arXiv:2412.21187.

Yanda Chen, Joe Benton, Ansh Radhakrishnan, et al. 2025. Reasoning models don't always say what they think. *Preprint*, arXiv:2505.05410.

DeepSeek-AI, Daya Guo, Dejian Yang, et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *Preprint*, arXiv:2501.12948.

Sitong Fang, Wenjing Cao, Jiahao Li, et al. 2025. When slower isn't truer: Inverse scaling law of truthfulness in multimodal reasoning. *Preprint*, arXiv:2505.20214.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. *Preprint*, arXiv:1705.08500.

Chuan Guo, Geoff Pleiss, Yu Sun, et al. 2017. On calibration of modern neural networks. *Preprint*, arXiv:1706.04599.

Jie Huang, Xinyun Chen, Swaroop Mishra, et al. 2024. Large language models cannot self-correct reasoning yet. *Preprint*, arXiv:2310.01798.

Renren Jin, Pengzhi Gao, Yuqi Ren, et al. 2025. Revisiting entropy in reinforcement learning for large reasoning models. *Preprint*, arXiv:2511.05993.

Saurav Kadavath, Tom Conerly, Amanda Askell, et al. 2022. Language models (mostly) know what they know. *Preprint*, arXiv:2207.05221.

Polina Kirichenko, Mark Ibrahim, Kamalika Chaudhuri, et al. 2025. Abstentionbench: Reasoning llms fail on unanswerable questions. *Preprint*, arXiv:2506.09038.

Romain Lacombe, Kerrie Wu, and Eddie Dilworth. 2025. Don't think twice! over-reasoning impairs confidence calibration. *Preprint*, arXiv:2508.15050.

Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. Teaching models to express their uncertainty in words. *Preprint*, arXiv:2205.14334.

Evan Miller. 2024. Adding error bars to evals: A statistical approach to language model evaluations. *Preprint*, arXiv:2411.00640.

Niklas Muennighoff, Zitong Yang, Weijia Shi, et al. 2025. s1: Simple test-time scaling. *Preprint*, arXiv:2501.19393.

Mario Sanz-Guerrero, Manuel Mager, and Katharina von der Wense. 2026. Large language models are overconfident in their own responses. *Preprint*, arXiv:2606.03437.

Rose Sathyanathan, Kinshuk Vasisht, and Danish Pruthi. 2026. Evaluating reasoning models for queries with presuppositions. *Preprint*, arXiv:2605.03050.

Rulin Shao, Shuyue Stella Li, Rui Xin, et al. 2025. Spurious rewards: Rethinking training signals in rlvr. *Preprint*, arXiv:2506.10947.

Zhihong Shao, Peiyi Wang, Qihao Zhu, et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *Preprint*, arXiv:2402.03300.

Mrinank Sharma, Meg Tong, Tomasz Korbak, et al. 2023. Towards understanding sycophancy in language models. *Preprint*, arXiv:2310.13548.

Parshin Shojaee, Iman Mirzadeh, Keivan Alizadeh, et al. 2025. The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity. *Preprint*, arXiv:2506.06941.

Charlie Snell, Jaehoon Lee, Kelvin Xu, et al. 2024. Scaling llm test-time compute optimally can be more effective than scaling model parameters. *Preprint*, arXiv:2408.03314.

Gaurav Srivastava, Aafiya Hussain, Sriram Srinivasan, et al. 2025. Do llms overthink basic math reasoning? benchmarking the accuracy-efficiency tradeoff in language models. *Preprint*, arXiv:2507.04023.

Yang Sui, Yu-Neng Chuang, Guanchu Wang, et al. 2025. Stop overthinking: A survey on efficient reasoning for large language models. *Preprint*, arXiv:2503.16419.

Katherine Tian, Eric Mitchell, Allan Zhou, et al. 2023. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. *Preprint*, arXiv:2305.14975.

Miles Turpin, Julian Michael, Ethan Perez, et al. 2023. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. *Preprint*, arXiv:2305.04388.

David Wadden, Shanchuan Lin, Kyle Lo, et al. 2020. Fact or fiction: Verifying scientific claims. *Preprint*, arXiv:2004.14974.

Ziqi Wang, Xingzhou Lou, Meiqi Wu, et al. 2026. Calibration-aware policy optimization for reasoning llms. *Preprint*, arXiv:2604.12632.

Bingbing Wen, Jihan Yao, Shangbin Feng, et al. 2024. Know your limits: A survey of abstention in large language models. *Preprint*, arXiv:2407.18418.

Yang Xiang, Yixin Ji, Ruotao Xu, et al. 2026. When is thinking enough? early exit via sufficiency assessment for efficient reasoning. *Preprint*, arXiv:2604.06787.

Miao Xiong, Zhiyuan Hu, Xinyang Lu, et al. 2023. Can llms express their uncertainty? an empirical evaluation of confidence elicitation in llms. *Preprint*, arXiv:2306.13063.

Haoming Xu, Ningyuan Zhao, Yunzhi Yao, et al. 2026. Illusions of confidence? diagnosing llm truthfulness via neighborhood consistency. *Preprint*, arXiv:2601.05905.

Chenlong Yin, Zeyang Sha, Shiwen Cui, et al. 2025. The reasoning trap: How enhancing llm reasoning amplifies tool hallucination. *Preprint*, arXiv:2510.22977.

Yang Yue, Zhiqi Chen, Rui Lu, et al. 2025. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model? *Preprint*, arXiv:2504.13837.

Muru Zhang, Ofir Press, William Merrill, Alisa Liu, and Noah A. Smith. 2023. How language model hallucinations can snowball. *Preprint*, arXiv:2305.13534.

Wenbo Zhang, Hengrui Cai, and Wenyu Chen. 2025. Beyond the singular: Revealing the value of multiple generations in benchmark evaluation. *Preprint*, arXiv:2502.08943.

Shu Zhou, Rui Ling, Junan Chen, et al. 2026. When more thinking hurts: Overthinking in llm test-time compute scaling. *Preprint*, arXiv:2604.10739.